

Linear Models with Random Effects

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Why Random Effects

- In some experimental situations, the factor levels are chosen at random from a larger population.
- The experimenter wishes to draw conclusions about the entire population, not just the levels that were used in the experimental design.
- We care about **variability**, not individual effects of some factors

Example 1: Machine-to-machine variability in manufacturing

- A factory wants to compare **3 cutting speeds** for a machining process. The experiment is run on **6 machines** chosen from a large fleet of similar machines. Each machine runs all 3 speeds, with repeated measurements.
 - Response: surface roughness
 - Treatment of interest: cutting speed
 - Nuisance: machine
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- If the 6 machines are the only machines we care about, we could treat machine as fixed.
 - If the 6 machines are just a sample from many possible machines, then machine should be random.

Model

$$Y_{ij} = \mu + \tau_i + b_j + \epsilon_{ij}$$

where

- $i = 1, 2, 3$ indexes cutting speed
- $j = 1, \dots, 6$ indexes machine
- τ_i is the fixed treatment effect of speed
- $b_j \sim N(0, \sigma_b^2)$ is the random machine effect
- $\epsilon_{ij} \sim N(0, \sigma^2)$

Interpretation

- τ_i : average effect of speed across the machine population
- σ_b^2 : machine-to-machine variability
- σ^2 : within-machine measurement noise

Sometimes with replication:

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}$$

- **Why random effect is appropriate**

Because the machines are not scientifically important one by one. We do not care about “Machine 4 versus Machine 5” as a substantive conclusion. We care about:

- the variability across machines
- whether the speed effect is robust across the machine population
- what to expect on a future machine not in the study

- **What fixed effect model would miss**

If we fit machine as fixed, we can adjust for machine differences, but we do not get the right population interpretation:

- inference becomes conditional on these 6 machines
- no direct variance component for machine population heterogeneity
- no clean statement about future machines

Example 2: Students nested within classrooms

- A school district compares **2 teaching methods**. The experiment involves **20 classrooms**, each taught by a teacher. Students within the same classroom are measured on end-of-term test score.
- Response: student test score
- Treatment: teaching method
- Grouping factor: classroom
- Students in the same classroom tend to be more similar because they share: teacher, environment, peer group, schedule, etc.

Model

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}$$

where:

- $i = 1, 2$: teaching method
- $j = 1, \dots, J$: classroom
- $k = 1, \dots, n_j$: student within classroom
- τ_i : **fixed effect** of teaching method
- $b_j \sim N(0, \sigma_b^2)$: **random classroom effect**
- $\epsilon_{ijk} \sim N(0, \sigma^2)$: residual

Example 3: Clinical trials

- Patients sampled from a population
- Repeated measurements per patient

Model:

Patient: random effect

Treatment: fixed effect

Fixed vs Random Effects (Conceptual)

Aspect	Fixed Effect	Random Effect
Levels	Specific, chosen	Random sample
Goal	Estimate each level	Estimate variance
Inference	About these levels	About population
Parameters	Deterministic	Random variables

Model Specification

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}, \quad i = 1, \dots, a, \quad j = 1, \dots, b, \quad k = 1, \dots, n$$

with constraints and assumptions

$$\sum_{i=1}^a \tau_i = 0,$$

$$b_j \stackrel{iid}{\sim} N(0, \sigma_b^2), \quad \epsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2), \quad b_j \perp \epsilon_{ijk}.$$

Here:

- μ is the grand mean
- τ_i is the fixed effect of treatment i
- b_j is the random effect for block, classroom, subject, batch, machine, and so on
- ϵ_{ijk} is the within-unit error

Mean, Variance, Covariance

Because $E(b_j) = 0$ and $E(\epsilon_{ijk}) = 0$,

$$E(Y_{ijk}) = \mu + \tau_i.$$

So the mean depends only on treatment. The random effect does not change the mean structure. It changes the **covariance structure**.

Marginal variance

$$\text{Var}(Y_{ijk}) = \text{Var}(b_j) + \text{Var}(\epsilon_{ijk}) = \sigma_b^2 + \sigma^2.$$

Covariance for two observations in the same group j

If two observations share the same j , say Y_{ijk} and $Y_{i'jk'}$ with $(i, k) \neq (i', k')$, then

$$\text{Cov}(Y_{ijk}, Y_{i'jk'}) = \text{Cov}(b_j + \epsilon_{ijk}, b_j + \epsilon_{i'jk'}) = \sigma_b^2.$$

Covariance for two observations from different groups $j \neq j'$

$$\text{Cov}(Y_{ijk}, Y_{i'j'k'}) = 0.$$

Observations in the same group are correlated, and observations in different groups are independent.

Correlation Structure

The within-group correlation is

$$\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}.$$

This is the **intraclass correlation coefficient**, ICC.

Interpretation:

- $\rho \approx 0$: little within-group dependence
- ρ large: observations in the same group are very similar

This is one of the most important interpretations of the random effect.

Matrix form

Write the model as

$$\mathbf{Y} = X\beta + Zb + \epsilon,$$

where

- $\beta = (\mu, \tau_1, \dots, \tau_a)^\top$ after suitable coding
- $b = (b_1, \dots, b_b)^\top$
- $b \sim N(0, \sigma_b^2 I_b)$
- $\epsilon \sim N(0, \sigma^2 I)$

Then

$$\text{Var}(\mathbf{Y}) = V = Z \sigma_b^2 I_b Z^\top + \sigma^2 I.$$

This is the standard mixed model covariance matrix.

If each group j contains m observations, then each within-group covariance block is

$$V_j = \sigma^2 I_m + \sigma_b^2 J_m,$$

where J_m is the $m \times m$ matrix of ones.

A Toy Example

Experimental data:

obs	treatment	group	Y
1	A	1	10
2	B	1	12
3	A	2	9
4	B	2	11

Model

$$Y_{ijk} = \mu + \tau_i + b_j + \epsilon_{ijk}$$

$$\mathbf{Y} = \begin{bmatrix} 10 \\ 12 \\ 9 \\ 11 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}.$$

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \quad \mathbf{b} \sim N(0, \sigma_b^2 I_2), \quad \epsilon \sim N(0, \sigma^2 I_4).$$

model is

$$\mathbf{Y} = X\beta + Z\mathbf{b} + \epsilon.$$

How to get the covariance matrix V

Because $X\beta$ is fixed,

$$\text{Var}(\mathbf{Y}) = \text{Var}(Z\mathbf{b} + \epsilon).$$

Since $\mathbf{b}$ and ϵ are independent,

$$\text{Var}(\mathbf{Y}) = \text{Var}(Z\mathbf{b}) + \text{Var}(\epsilon).$$

Step 1: compute $ZZ^\top$

$$Z = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

so

$$ZZ^\top = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

This matrix is extremely important.

Its meaning is:

- entry = 1 if two observations are in the same group
- entry = 0 if they are in different groups

Step 2: plug into V

$$V = \sigma_b^2 \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} + \sigma^2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

So

$$V = \begin{bmatrix} \sigma_b^2 + \sigma^2 & \sigma_b^2 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 + \sigma^2 & 0 & 0 \\ 0 & 0 & \sigma_b^2 + \sigma^2 & \sigma_b^2 \\ 0 & 0 & \sigma_b^2 & \sigma_b^2 + \sigma^2 \end{bmatrix}.$$

How to interpret this matrix

Each diagonal entry is

$$\text{Var}(Y) = \sigma_b^2 + \sigma^2.$$

For observations in the same group, such as obs 1 and obs 2,

$$\text{Cov}(Y_1, Y_2) = \sigma_b^2.$$

For observations in different groups, such as obs 1 and obs 3,

$$\text{Cov}(Y_1, Y_3) = 0.$$

So the covariance matrix immediately shows:

- same group $\Rightarrow$ correlated
- different groups $\Rightarrow$ independent

Estimation of fixed effects when variance components are known

If V were known, the generalized least squares estimator is

$$\hat{\beta} = (X^{\top} V^{-1} X)^{-1} X^{\top} V^{-1} \mathbf{Y}.$$

Its covariance matrix is

$$\text{Var}(\hat{\beta}) = (X^{\top} V^{-1} X)^{-1}.$$

This is the correct estimator for μ and τ_i under correlated observations.

When variances are unknown

- Two-step estimation

1. Estimate variance components

$$\sigma_b^2, \sigma^2$$

$$\hat{\sigma}^2 = MS_E$$

$$\hat{\sigma}_b^2 = \frac{MS_{\text{group}} - MS_E}{n}$$

2. Plug into $\hat{V}$ and use GLS:

$$\hat{\beta} = (X^\top \hat{V}^{-1} X)^{-1} X^\top \hat{V}^{-1} Y$$

- Iterative maximum likelihood